

A Smart Attendance Monitoring Web Application Using RFID, Face Recognition, and IoT-Based Analytics for DYCI

1. RFID-Based Attendance Systems

This study presents an RFID-based attendance monitoring system designed to automate student attendance in schools. The system significantly reduced manual attendance recording and improved efficiency. However, the study relied solely on RFID cards, which can be misused through card sharing. This limitation supports the need for integrating face recognition in the proposed system for DYCI.

2. Face Recognition for Attendance Monitoring

Patel et al. (2020) developed an attendance system using facial recognition technology to improve identity verification accuracy. The system minimized proxy attendance but required stable lighting and camera conditions. The proposed DYCI system enhances this approach by combining face recognition with RFID and IoT-based analytics for higher reliability.

3. IoT-Based Smart Monitoring Systems

This literature discusses IoT architectures and applications, including smart monitoring systems in educational environments. It highlights how IoT enables real-time data collection and system integration. However, the study does not focus on attendance applications, which the proposed DYCI system specifically addresses.

4. Web-Based Attendance Management Systems

Rahman et al. (2022) developed a web-based attendance system that allowed administrators to view attendance records online. While the system improved accessibility, it depended on manual input. This gap supports the need for automated data collection using RFID, face recognition, and IoT analytics as proposed in this study.

5. Analytics-Driven Educational Systems

This literature emphasizes the importance of analytics in educational decision-making by transforming raw data into actionable insights. However, it does not specify how attendance data is collected. The proposed DYCI system bridges this gap by integrating IoT-generated attendance data with analytics for improved academic monitoring.

<https://jcrinn.com/index.php/jcrinn/article/view/507>

<https://ijcse.isroset.org/index.php/j/article/view/2435>

<https://arxiv.org/abs/2308.02591>

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<https://researchvision.us/index.php/smartify/article/view/85>